

B

Study Buddy

GCSE Maths practice that adapts to what you get wrong.

Continue with [PROVIDER]

or

School or personal email

you@example.com

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GCSE Maths

S

Evening, [NAME].
Twenty minutes on quadratics?

PICK UP WHERE YOU LEFT OFF

Solving quadratics by factorising

4 of 10

Resume

BUDDY NOTICED

You've slipped on the sign twice with (x - a). Three quick questions on that are queued.

Do the three now --

Quadratics

See progress

Solving by factorising

72% correct

Today

Practice

Progress

Notes

GCSE Maths

B

Let Buddy choose for me

Quadratics

Solving by factorising

72% correct

Coming Later

Simultaneous equations

Inequalities

Sequences

Today

Practice

Progress

Notes

<

Quadratics - Solving by factorising

How many questions would you like to solve?

5 questions

10 questions

15 questions

20 questions

Start 10 questions

X

4 / 10

Focus: signs in brackets

Solve by factorising.

$x^2 - 5x + 6 = 0$

Your answer

x = 2, x = -3

Check answer

Your working

Listening - 0:14

"Two numbers that multiply to six and add to minus five... minus two and minus three. So x minus two, x minus three. Then x equals two, and x equals minus three."

X

5 / 10

Focus: signs in brackets

Solve by factorising.

$x^2 - 7x + 12 = 0$

Your answer

Type your answer

Check answer

X

4 / 10

Focus: signs in brackets

Solve by factorising.

$x^2 - 5x + 6 = 0$

Your answer

x = 2, x = -3

Check answer

Your working

Recorded - 0:14

"Two numbers that multiply to six and add to minus five... minus two and minus three. So x minus two, x minus three. Then x equals two, and x equals minus three."

Wrong, try again

Try again

Hint

Skip

X

4 / 10

Focus: signs in brackets

Solve by factorising.

$x^2 - 5x + 6 = 0$

Your answer

x = 2, x = -3

Check answer

Your working

Recorded - 0:14

"Two numbers that multiply to six and add to minus five... minus two and minus three. So x minus two, x minus three. Then x equals two, and x equals minus three."

Wrong, try again

Sign error

Check the sign when you solve each bracket. If x - 3 = 0, what does x equal?

Try again

Skip

X

4 / 10

Focus: signs in brackets

Solve by factorising.

$x^2 - 5x + 6 = 0$

Your answer

x = 2

Check answer

Wrong, try again

Stopped at one root

You've found one solution, but this equation has two. Solve the other bracket too.

Try again

Skip

X

4 / 10

Focus: signs in brackets

Solve by factorising.

$x^2 - 5x + 6 = 0$

Your answer

x = 1, x = 6

Check answer

Wrong, try again

Try again

Skip

X

4 / 10

Incorrect

YOU WROTE

x = 2, x = -3

CORRECT

x = 2, x = 3

Your factorising was spot on: (x - 2)(x - 3) = 0. The slip came at the last step.

If x - 3 = 0, add 3 to both sides, so x = 3.

What Buddy is doing next

1 Logged a sign slip (2nd this week).

2 Next: both roots positive, to check the fix sticks.

From your revision notes

"Set each bracket equal to zero..."

Explain again

Next

Progress

Quadratics

Solving by factorising

72% correct

Recurring mistakes

Sign flip on the final step

2 sessions - being practised

Sign error in the second bracket

Once - watching

This week

minutes / day

M T W T F S S

Today

Practice

Progress

Notes

Quadratics - Solving by factorising

Revision notes

Make it equal zero first

Move every term to one side before you factorise.

$x^2 - 5x = -6 \rightarrow x^2 - 5x + 6 = 0$

Find the pair

Two numbers that multiply to the number on its own and add to the number in front of x.

$-2 \times -3 = 6$ and $-2 + -3 = -5$

Write the brackets

$(x - 2)(x - 3) = 0$

USED IN YOUR LAST FEEDBACK

Set each bracket equal to zero

Then solve each one on its own.

$x - 2 = 0 \rightarrow x = 2$

$x - 3 = 0 \rightarrow x = 3$

Check both answers

Put each one back into the equation. It should give zero.

$2^2 - 5 \times 2 + 6 = 0$ ✓

Today

Practice

Progress

Notes